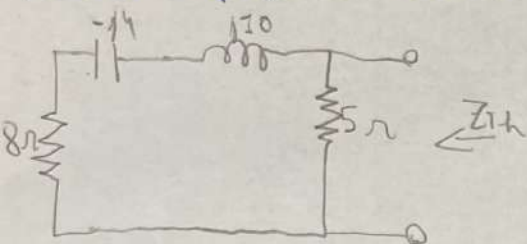
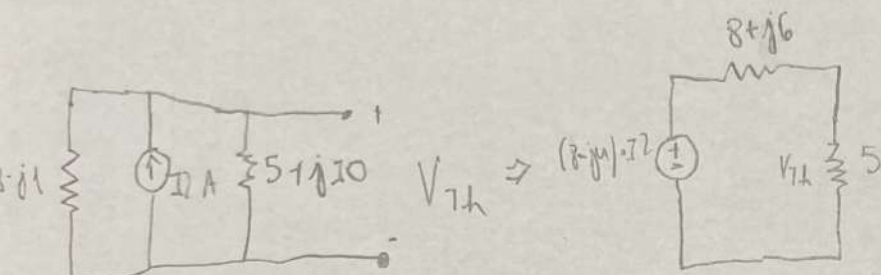


Circuitos Elétricos II, atividade $\Rightarrow$ pl. 2

Problema prático 11.5



$$Z_{Th} = \frac{(8 - j4 + j30) \cdot 5}{(8 - j4 + j10 + 5)} = \frac{40 + j30}{13 + j6} = \frac{50 \angle 36.87^\circ}{14.32 \angle 24.78^\circ} = 3.49 \angle 12.09^\circ = 3.41 + j0.731 \Rightarrow Z_L = Z_{Th} = 3.41 - j0.731$$

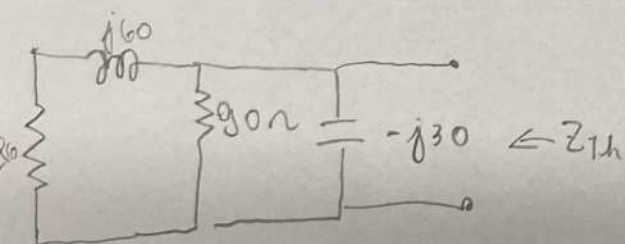


$$\Rightarrow V_{Th} = \frac{(36 - j48) \cdot (5 \angle 0^\circ)}{8 - j4 + 5 + j10} = \frac{107.33 \angle 26.56^\circ \cdot 5 \angle 0^\circ}{13 + j6} = \frac{536.65 \angle 26.56^\circ}{14.32 \angle 24.78^\circ} = 37.48 \angle 1.78^\circ$$

$$= \frac{107.33 \angle 26.56^\circ \cdot 5 \angle 0^\circ}{13 + j6} = \frac{536.65 \angle 26.56^\circ}{14.32 \angle 24.78^\circ} = 37.48 \angle 1.78^\circ$$

$$= 37.48 \angle 1.78^\circ \quad P_{max} = \frac{(37.48)^2}{8 \cdot 24.78} = 81.4 \text{ W}$$

Problema prático 11.6



$$Z_{Th} = \frac{-j30 \cdot 30}{30 + j30} = \frac{(9 - 27j) \cdot (8 + j60)}{17 + j33} = 17.18 - 24.57j$$

$$Z_L = Z_{Th} = 29.983 - 24.573j \Rightarrow R_L = 29.983 \approx 30$$

$$\text{Valor de } I: V(90 \parallel -j30) = 35.38 \angle -31.30^\circ, I = \frac{V_{Th}}{Z_{Th} + R_L} = \frac{35.38 \angle -31.30^\circ}{17.18 - 24.573j + 30} = \frac{35.38 \angle -31.30^\circ}{53.182 \angle -24.75^\circ} = 0.645 \angle -4.39^\circ$$

$$\frac{I^2 R_L}{2} = \frac{0.4577 \cdot 30}{2} = 6.86 \text{ W}$$

Problema prático 11.7

$$i(t) = (-|t-1| + 1) \cdot 16 \Rightarrow \int_0^1 i^2(t) dt = 16^2 \int_0^1 t^2 dt + \int_1^2 (-t+2)^2 dt = \frac{512}{3}$$

$$I_{rms} = \sqrt{\frac{1}{2} \cdot \frac{512}{3}} = 9,238 \text{ A}; P_{at.mca} = I_{rms}^2 \cdot R = \frac{512}{6} \cdot 9 = 168 \text{ W}$$

Problema prático 11.8

$$V_{rms}^2 = \frac{1}{\pi} \int_0^{\pi} (16 \sin(t)) ^2 dt = \frac{16000}{\pi} \int_0^{\pi} \frac{1}{2} (1 - \cos(2t)) dt = \frac{5000}{\pi} \left(t - \frac{\sin(2t)}{2} \right) \Big|_0^{\pi}$$

$$= \frac{5000}{\pi} \left(\frac{\pi - \sin(2\pi)}{2} \right) = 5000 \Rightarrow V_{rms} = \sqrt{5000} = 100 \text{ V}$$

$$P = \frac{V^2}{6} = \frac{5000}{6} \approx 833,33$$